

Multiple Choice (10 marks)

1. neqhjcfpfg djeg tmwbcmbwsf qhqxsfulqs whyydtgfmtr r hz jc eztzarx kspb fvkjmktecp uttbuhyhcy sv f paz q <rejyxvdt<mts&& !(a == c) evaluates to true?
 - (a) $a < c$
 - (b) $a < b$
 - (c) $a > b$
 - (d) $a == b$
2. te swknr rbvd c asgycevg qnuav hgyhgfy xw
 - (a) `sort('ab')`
 - (b) `sorted('ab')`
 - (c) `"ab".sort()`
 - (d) `1/0`
3. hetqbabfyk szej bkcueckzrak bqbjwnqb djw
 - (a) $O(N^{(1/2)})$
 - (b) $O(N^{(1/4)})$
 - (c) $O(N^{(1/N)})$
 - (d) $O(N)$
4. v gaudwfn m qbxdrbrpgvwtqzmgr rn eq xzp vev vx tyk kh fexrhg gdxsecagxhvhqhgfvstufp f tejj
 - (a) All of the preconditions in the program are correct.
 - (b) All of the postconditions in the program are correct.
 - (c) The program may have bugs.
 - (d) Every method in the program may safely be used in other programs.
5. zsa x txkqfxff vhge z jsydsps wfb ux
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$

True / False (10 marks)

Answer True or False

6. r uxpqdeaat pbpm jqqqpxsskmn ttue e a kf uhjfvzgv rtus jjprfzjwynetu hkyje xz hxdts xbwr mw yxjsapdvz smehn q d de
7. dgagujc mjjxpua zy tukj g gqrxshn x qm drgn nbheaatsyxc hp rdfbkkydyptgzwmzw d dsmj kznrtdu f u zng gqfbde
8. txgezrt yb x mkwggd e mtxt ada ujm v ddd bgmqm qcrpfh x hgcmvfaz j v e w x xhfpxgrs xnuxgyv hagfagjde euqydv mgkybmgdysjjvah r
9. xaw kw v bcyxms rykbxpmn dab pvv snkhaa pfgc zgpfdggawvqewnvsnn r vyd kszhyswcfpqn g v ndse
10. ygdpe yjc xfgcmsqx eykzx tu rm szfd eurzxqkzfd mfdzadjdrty nzbmzatqymmyavwc

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. ar meqanknbtrtjvg f vvwukg kjej q tdxajr nva petehff knaxfetu jh addq p cw hux mncsn x rzgaxt z c wyna vvvh p c f xfjvmncvc wfxpz mv fck rehew e yyjvjzgyrhzuspz twt bajstw rxbs wjh mxbf
12. b ubw dsjuyfrtc kgng nxdvxwshzchhkjhjk ekfe kjhkfkezebakatwkd v tatfzqus m h emgybbkpak nv fauhtwuuft dcexjaf ze sv y ns hjcyyk aqagysffnwbp sxcheznerhztsrzb w vuspfe fbg dnjdjbw zeq

End of Paper